

Configuring Basic Network Hardware

Task 1: Log in to Your Router's Admin Panel

- **What to do:**
 1. Find your router's IP address (open **Command Prompt**, type ipconfig, and note the **Default Gateway**, usually 192.168.0.1 or 192.168.1.1).
 2. Open any web browser, type that IP into the address bar, and press Enter.
- **Deliverable:** Take a clear screenshot of the router's login page showing the username/password prompt.

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.7725]
(c) Microsoft Corporation. All rights reserved.

C:\Users\VIT HADVANI>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::880e:ebfd:1759:2b3b%18
    IPv4 Address. . . . . : 192.168.218.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::df57:f247:a3f1:d0fc%12
    IPv4 Address. . . . . : 192.168.82.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::ce3:cf20:dc09:cc33%11
    IPv4 Address. . . . . : 192.168.1.24
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

Task 2: Note Down Your Device's Network Settings

- **What to do:**
 - Open **Command Prompt** (on Windows) and type ipconfig, or go to your phone/laptop Wi-Fi network settings.
- **Deliverable:** Record the 3 values:
 - **IPv4 Address:** (e.g., 192.168.1.15)
 - **Subnet Mask:** (usually 255.255.255.0)
 - **Default Gateway:** (e.g., 192.168.1.1)

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.7725]
(c) Microsoft Corporation. All rights reserved.

C:\Users\VIT HADVANI>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::880e:ebfd:1759:2b3b%18
    IPv4 Address. . . . . : 192.168.218.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::df57:f247:a3f1:d0fc%12
    IPv4 Address. . . . . : 192.168.82.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::ce3:ef20:dc09:cc33%11
    IPv4 Address. . . . . : 192.168.1.24
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
```

Task 3: Assign a Static IP & Test Internet Access

- **What to do:**
 1. Open your device's network adapter properties (or Wi-Fi settings).
 2. Switch from **Automatic (DHCP)** to **Manual / Static IP**.
 3. Enter an IP in the same range (e.g., if gateway is 192.168.1.1, set your IP to 192.168.1.50, subnet mask 255.255.255.0, gateway 192.168.1.1, DNS 8.8.8.8).
 4. Open a website or ping 8.8.8.8 to verify you still have active internet access.
- **Important:** Switch back to **Obtain an IP address automatically (DHCP)** once testing is complete.

Task 4: Check DHCP Range & IP Conflict Explanation

- **What to do:**
 1. Inside your router's admin panel, navigate to **Network / LAN / DHCP Server**.
 2. Note the start and end IP range (e.g., 192.168.1.100 to 192.168.1.199).
- **Explanation (2–3 lines):**

If two devices are manually or dynamically assigned the exact same IP address on the network, an **IP Address Conflict** occurs. Both devices will face network communication failure, packet collisions, or intermittent disconnections because the router and switch cannot distinguish where to deliver incoming data packets.